MIL-C-44061A
30 September 1983
SUPERSEDING
MIL-C-44061
29 December 1982

MILITARY SPECIFICATION

CHICKEN LOAF, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers thermostabilized chicken loaf in flexible pouches for use as a component of the meal, ready-to-eat, individual.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

C-D-221	-	Dessert Powders and Gelatin, Plain, Edible
EE-S-631	-	Spices, Ground and Whole, and Spice Blends

MILITARY

MIL-P-44073	-	Packaging and Thermoprocessing of Foods in Flexible Pouches

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8905

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks, and Sugars Used in the Preparation of Canned Foods for the Armed Forces

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Poultry Products Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Regulations Governing the Inspection of Egg and Egg Products

(Application for copies should be addressed to the Poultry Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4, 6.1 and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Chicken. The chicken shall be prepared from chilled or frozen ready-to-cook fowl. It shall be in excellent condition at the time of use. Time from date of slaughter to incorporation into the finished product shall not exceed 120 days. Chickens chilled or frozen (ready-to-cook) shipped between plants shall be accompanied by a USDA Grading Branch Certificate.

3.2.1.1 Handling and storage of raw chicken. Handling and storage of raw chicken prior to preparation into the finished product shall be in accordance with the following:

a. Chicken processed on the same day as boning and trimming shall not exceed 38°F at any time prior to processing.

b. Holding in the fresh chilled state is permitted provided that the chicken is maintained in the temperature range of 28° to 36°F (inclusive) at all times prior to cooking.

c. Holding the raw chicken in the frozen state is permitted, provided that the chicken is subjected to freezing within 72 hours after acceptance and stored at 0°F or lower at all times.

d. Meat shall not be frozen more than once prior to cooking.

3.2.2 Rendered chicken fat. The rendered chicken fat shall have a clean yellow color and a characteristic mild chicken-fat odor and flavor. The rendered chicken fat may be obtained from chicken broth or by rendering chicken skin or abdominal fat (leaf fat) from chickens inspected and passed by the FSIS of the USDA. Time from rendering to incorporation into finished product shall not exceed 72 hours and the temperature shall be 40°F or below (there shall be no evidence of freezing).

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The rendered fat may be protected by the types and amounts of antioxidants approved by the FSIS of the USDA. When frozen rendered chicken fat is used, it shall not be stored for more than 75 days at a temperature of 0°F or lower from time of initial freezing until use in the finished product. Frozen rendered chicken fat shall have a peroxide value not to exceed 6 me/kg, a free fatty acid value not to exceed 0.75 percent oleic and moisture content not to exceed 0.25 percent.

3.2.3 Cracker meal. Cracker meal shall be made from flour and water only. It shall be granular, free flowing and free from burnt, rancid, moldy, or musty odor and flavor. It shall comply with the requirements of MIL-STD-900.

3.2.4 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agent.

3.2.5 Monosodium glutamate. The product shall be odorless, free flowing and may have either a slightly sweet or slightly salty flavor. The crystals shall be white and translucent.

* 3.2.6 Egg whites. The egg whites may be liquid or frozen and shall have been processed in accordance with Regulations Governing the Inspection of Eggs and Egg Products. The egg whites shall contain no additives, and shall be in satisfactory condition, free of objectionable odors, such as musty, sour, or fruity, and free from foreign material. Liquid egg products shall be held at a temperature of 40°F or lower and shall not be held more than 72 hours from time of pasteurization until start of formulation of chicken loaf. Frozen egg whites shall be held at 10°F or lower and used within 120 days. Thawed, previously frozen, egg products shall be held at a temperature of 40°F or lower and shall not be held for more than 24 hours prior to use in processing.

3.2.7 Gelatin, dry. Dry gelatin shall conform to type I (plain, unflavored) of C-D-221 and shall be of the bloom specified in the formula (see 3.3.2).

3.2.8 White pepper. White pepper shall conform to type II of EE-S-631.

3.2.9 Chicken broth. The chicken broth shall be clear, fresh, canned or frozen, essentially fat free, with approximately 3.0 to 3.5 percent solids and may contain any combination of seasonings and flavor enhancers such as monosodium glutamate, disodiuminosinate or disodium guanylate. The chicken broth shall be produced in USDA inspected plants in accordance with USDA Regulations.

3.3 Preparation and processing of product.

3.3.1 Chicken preparation. The chicken shall be prepared as follows:

a. The feet, metacarpus-phlanges section of the wings, giblets, bones, coarse connective tissue, skin, blood clots, bruises, or discolored meat exceeding 0.3 inch in any dimension shall be excluded from the formulation.

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b. The deboned chicken meat shall consist of not less than 50 percent nor more than 60 percent white meat with the remaining percentage dark meat and shall be either water cooked, roasted, or steamed. Save broth or drippings for use in loaf preparation.

c. Cooking in casing with the proper proportions of light to dark meat is permissible. Rolls shall be formulated without skins.

d. The chicken meat shall be diced no greater than 3/4 inch in the longest dimension and not less than 1/2 inch in the shortest dimension.

e. Precooked, frozen, chicken dices of the same dimensions as specified above may be purchased from a commercial deboning operation and shall comply with the requirements of 3.2.1, and 3.3.1, items a, b and d. If frozen, precooked chicken is used, fresh, canned or frozen broth will have to be purchased.

* 3.3.1.1 Handling and storage of cooked chicken. Cooked chicken dices not prepared into the finished product on the same day as cooking shall be handled and stored in accordance with the following requirements:

a. Holding cooked chicken dices in the frozen state for not more than 14 days is permitted, provided that the chicken is placed in a freezer at 0°F or lower within 4 hours after cooking and frozen to 0°F within 8 hours after placement in the freezer.

b. Holding cooked chicken dices in the chilled or thawed state not more than 72 hours prior to final preparation of product is permitted, provided that the temperature of the chicken does not exceed 40°F at any time. There shall be no evidence of freezing or refreezing.

* 3.3.2 Chicken loaf preparation. Chicken loaf shall be prepared in accordance with the following formulation:

Ingredient	Percent by weight
Chicken, fowl, cooked diced	66.18
Egg whites	12.50
Chicken broth (3.0 - 3.5 percent solids)	10.00
Chicken fat	5.00
Cracker meal	5.00
Salt	0.50
Gelatin, dry, (275 bloom)	0.50
Monosodium glutamate	0.20
White pepper	0.12

a. The cracker meal, salt, monosodium glutamate and white pepper shall be premixed. The gelatin shall be added to the chicken broth.

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b. Frozen egg whites, if used, chilled or frozen chicken fat shall be tempered sufficiently to insure complete dispersion during the mixing process.

c. With the use of mechanical mixing, the dry premix, chicken broth - gelatin mix, egg whites, and chicken fat shall be added to the chicken dices in the order mentioned.

* 3.3.3 Pouch filling and processing. Each pouch shall be filled and processed such as to conform to the following requirements:

a. Filling, sealing, and thermoprocessing shall conform to the class 3 requirements of MIL-P-44073.

b. Temperature of the mixture at time of filling into pouches shall be 45°F or lower. The mixture may be held at 45°F or below for a maximum of 4 hours before filling into the pouches.

c. Average net weight shall be 5.0 ounces minimum.

d. No individual pouch shall contain less than 4.8 ounces of product.

e. The average drained weight of chicken loaf shall be 4.5 ounces minimum.

f. No individual pouch shall contain less than 4.3 ounces of drained chicken loaf.

g. The filled and sealed pouch shall be thermoprocessed until a sterilization value (F_o) of not less than 6 has been achieved.

* 3.4 Finished product. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not restricted to glass, paint, metal, dirt, insect, insect part, hair, wood, or paper.

b. There shall be no foreign odor or flavor such as, but not restricted to burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no foreign color such as, but not restricted to black, purple, or green.

d. There shall be no bone pieces measuring more than 0.3 inch in any dimension.

e. There shall be no cartilage or section of tendons or ligaments measuring more than 0.3 inch in any dimension.

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f. There shall be no pieces of skin, blood clot, or bruised or discolored meat exceeding 0.3 inch in any dimension.

g. Each pouch shall contain one unbroken chicken loaf.

h. There shall be no air pocket in the loaf exceeding 0.5 inch in any dimension.

i. Texture of chicken loaf shall not be dry, rubbery, or mushy.

j. The fat content shall not exceed 11 percent.

k. The salt content shall not exceed 1.0 percent.

3.4.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.5 Plant qualification. The poultry products component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this document. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

* 4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

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4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Chicken examination. Examination of chicken shall be made to determine conformance to the requirements of 3.2 and 3.2.1. Records of temperature and storage time shall be maintained. Nonconformance to one or more condition requirements or to temperature and storage requirements, reflected by actual examination or by records shall be cause for rejection of the involved quantity of product or the finished product made therefrom, as applicable.

4.5.1.2 Other ingredient examination. Conformance of ingredients to identity and condition requirements shall be determined by examination of pertinent labels, markings, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient lot or of any involved product.

4.5.1.3 Frozen chicken fat testing (when applicable). The frozen chicken fat shall be tested for the characteristics listed in table I. The sample shall be a 1-pound composite derived from five primary containers or from all containers if there are less than five containers in the ingredient lot. Any test failure shall be cause for rejection of the lot.

TABLE I. Frozen chicken fat tests

Characteristic	Requirement paragraph	Test method	Results reported to nearest
Peroxide value	3.2.2	Peroxide value method 1/	1 me/kg
Free fatty acid value	3.2.2	Free fatty acids method 1/	0.01 percent (as oleic acid)
Moisture	3.2.2	Vacuum oven method 1/	0.01 percent

1/ From the Oils and Fats chapter of Official Methods of Analysis of the Association of Official Analytical Chemists.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation and processing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or other valid documents shall be cause for rejection of the involved product.

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4.5.3 Filled and sealed pouch inspection. Inspection of filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-440

4.5.4 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 4.8 ounces sh be scored as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 0.1 ounce. In addition, the lot shall be rejected if the sample average net weight is less than 5.0 ounces.

* 4.5.5 Product inspection. The filled and sealed pouches shall be held for a minimum of 72 hours at room temperature (65° to 75°F) after completion of the thermoprocessing operation. The pouches shall be heated in boiling water, opened, and inspected for the defects indicated in table II. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundr units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE II. Product inspection 1/ 2/

Category		Defect
Major	Minor	
151		Presence of bone measuring more than 0.3 inch in any dimension.
152		Presence of cartilage or section of tendons or ligaments measuring more than 0.3 inch in any dimension.
153		Presence of skin, blood clot, or bruised or discolored meat measuring more than 0.3 inch in any dimension.
154		Drained weight of chicken loaf in a pouch is less than 4.3 ounces. 3/ 4/
155		Color not characteristic of product (materially darkened, scorched, or oxidized).
156		Texture of chicken loaf dry, rubbery, or mushy.

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TABLE II. Product inspection 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
	201	Broken loaf.
	202	Presence of air pocket measuring more than 0.5 inch in any dimension.

1/ Presence of any foreign material (e.g. glass, dirt, insect part, hair, wood, metal, etc.), foreign odor or flavor (e.g. burnt, scorched, stale, sour, rancid, musty, moldy, etc.), or foreign color shall be cause for rejection of the lot. (To be determined by a Meat and Poultry Inspection Operations employee.)

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

3/ To determine drained weight, the contents shall be poured into a U.S. standard 1/2 inch sieve in a manner that will distribute the product over the sieve without breaking the chicken loaf. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted slightly and allowed to drain at least 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

4/ If the sample average drained weight is less than 4.5 ounces, the lot shall be rejected.

4.5.6 Product testing. The product shall be tested for fat content as specified in 4.5.6.1 and for salt content as specified in 4.5.6.2. The lot size shall be expressed in pouches. The sample unit shall be the contents of one filled and sealed pouch. The sample size shall be the number of sample units indicated by inspection level S-2. Any test result not conforming to the applicable fat or salt content requirement specified in 3.4 shall be cause for rejection of the lot.

* 4.5.6.1 Fat content testing. The product shall be tested for fat content in accordance with the Fosslet method or the Crude Fat or Ether Extract method (using petroleum ether) in the Meat and Meat Products chapter of the Official Methods of Analysis of the Association of Analytical Chemists. The results shall be reported to the nearest 0.1 percent.

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4.5.6.2 Salt content testing. The product shall be tested for salt content in accordance with the Meat section in the Meat and Meat Products chapter of the Official Methods of Analysis of the Association of Analytical Chemists. The results shall be reported to the nearest 0.1 percent.

4.5.7 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44073.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the applicable requirements of MIL-P-44073.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.3).
c. Provisions for approved preproduction samples (see 3.4.1 and 6.3).

6.2 Award of contract. Award of contract for the product specified in this document will be limited to plants known to maintain the required sanitation conditions of 3.5.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, TS
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

Project No. 8905-0996

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-44061A	Chicken Loaf, Thermostabilized, For Meal, Ready-To-Eat

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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